Biochemistry Cumulative Exam

(Supplemental Information Attached)

A mouse is fed a diet including the amino acid alanine, but no carbohydrates, or fatty acids. Small amounts of the essential amino acids and the necessary vitamins and minerals are also included in the diet.

1. (40) Outline a metabolic pathway for the mouse to synthesize isocitrate on this diet using alanine as the <u>sole source</u> of carbon, and energy. How many alanine molecules must be used to provide enough carbon atoms, ATP, and NADH molecules to synthesize each molecule of isocitrate? Write a complete, balanced net equation for this process giving the numbers of ATP, NADH, $FADH_2$ and CO_2 generated.

2. (40) Outline a metabolic pathway for the mouse to synthesize glutamate on this diet using alanine as the <u>sole source</u> of carbon, nitrogen and energy. How many alanine molecules must be used to provide enough carbon atoms, nitrogen, ATP, and NADH molecules to synthesize each molecule of glutamate? Write a complete, balanced net equation for this process giving the numbers of ATP, NADH, $FADH_2$ and CO_2 generated.

3. (20) If alanine is labeled with ^{14}C at carbon 2, which carbon(s) of glutamate and isocitrate will first become labeled?

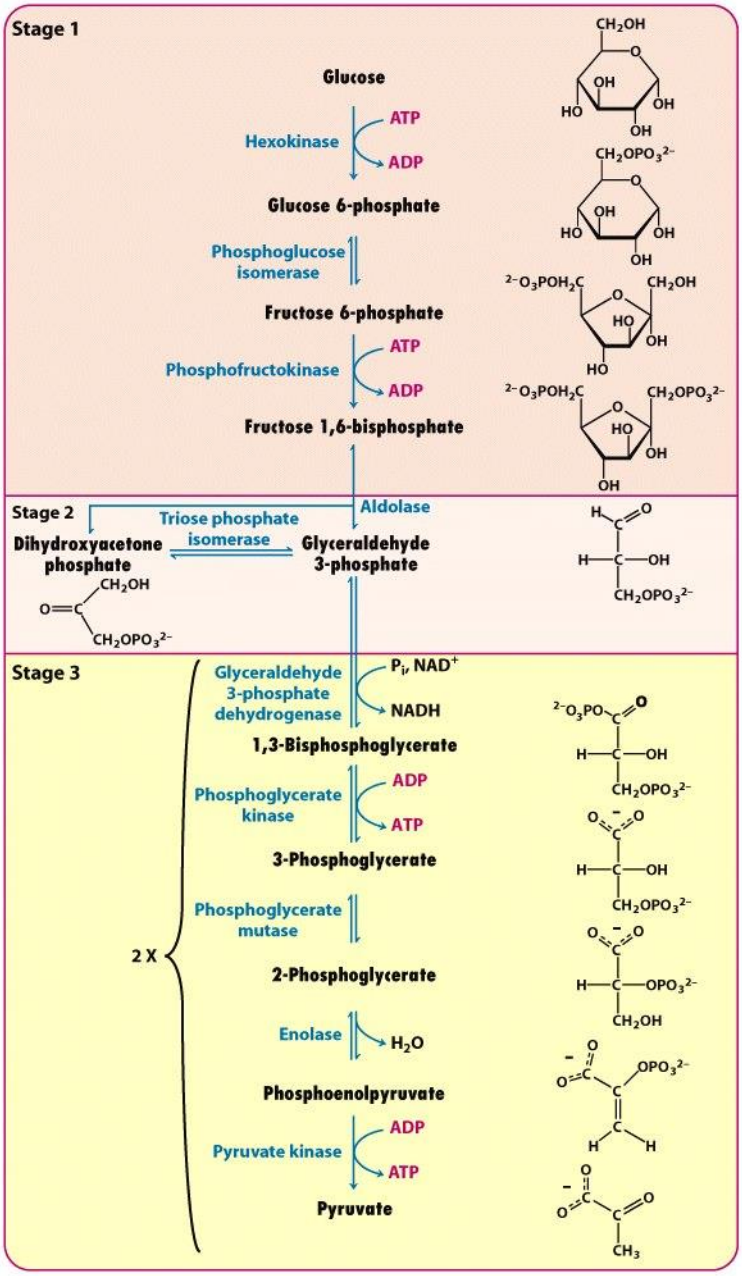

Figure 16-2
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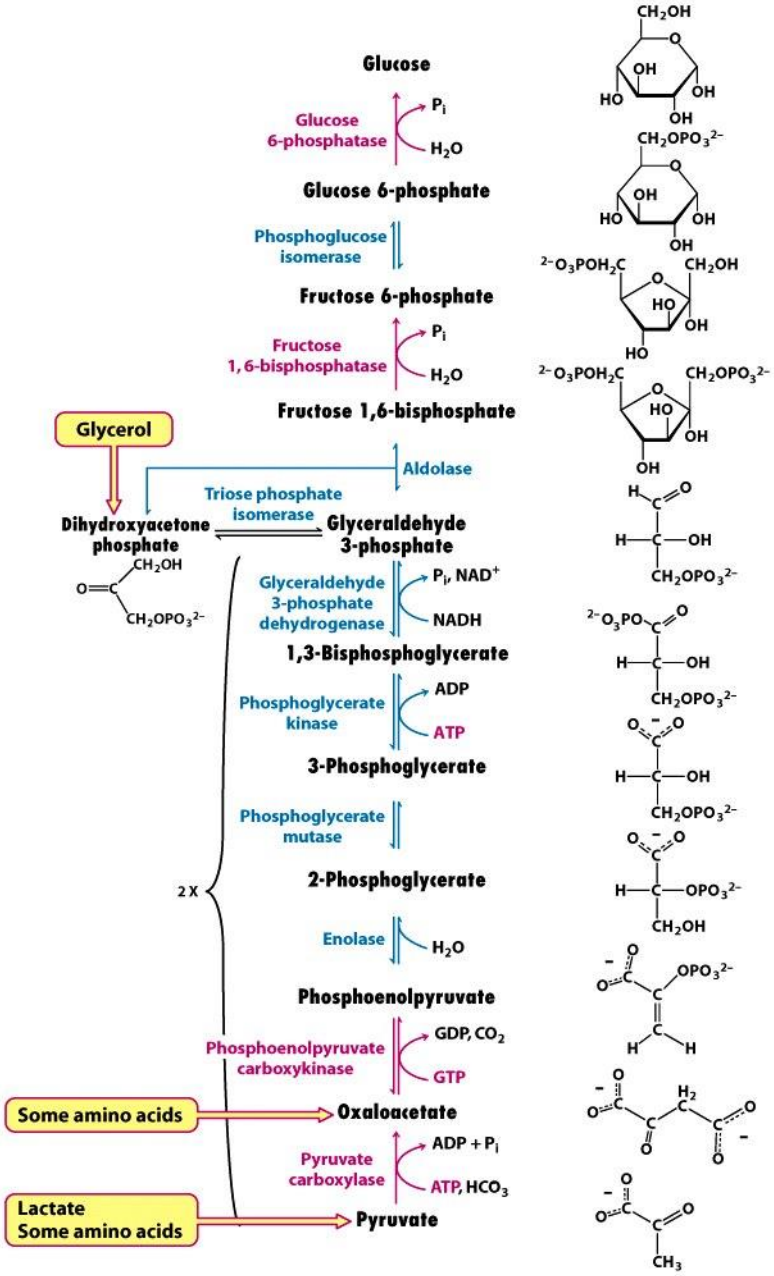

Figure 16-22
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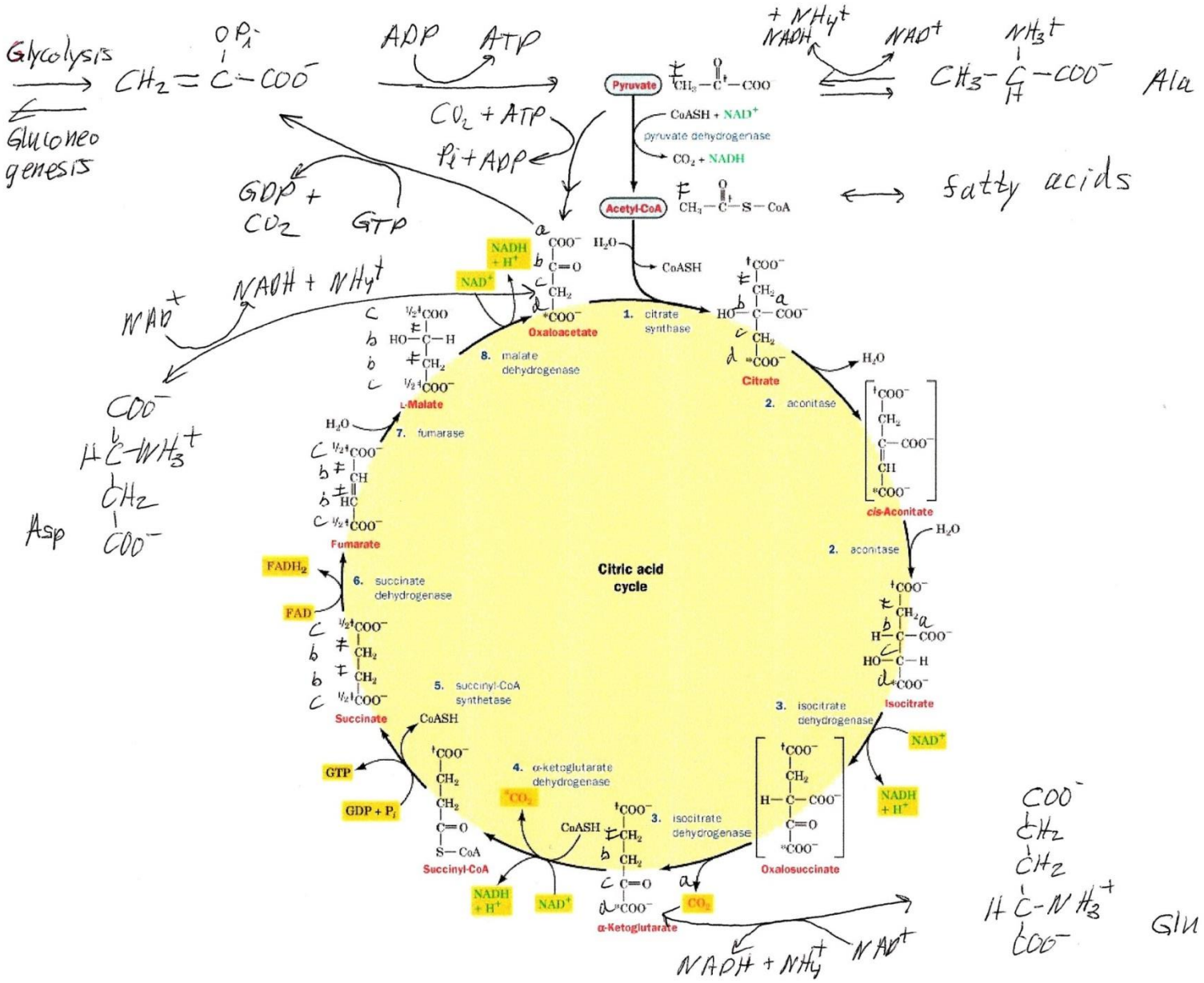
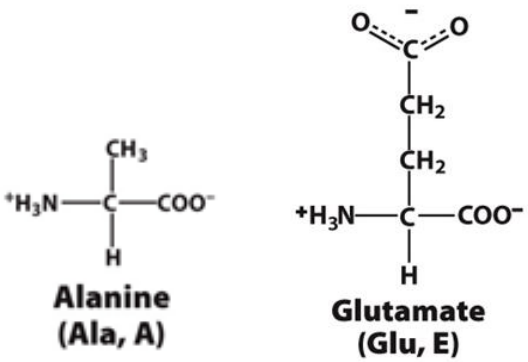
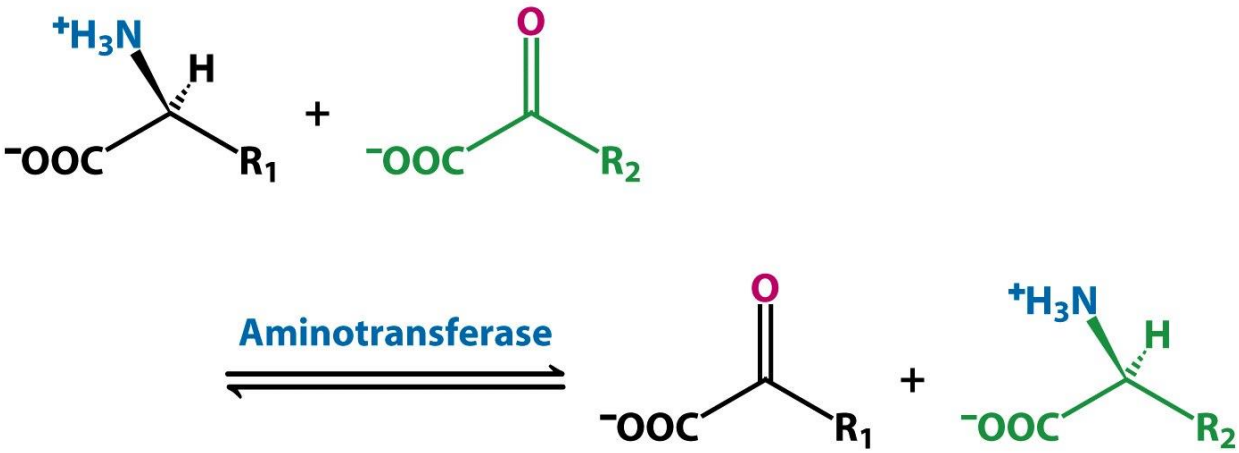

Glycolysis
Gluconeo genesis
ADP
ATP
Pyruvate
CO2 + ATP
Pi + ADP
GDP + CO2
GTP
CoASH + NAD+
pyruvate dehydrogenase
CO2 + NADH
Acetyl-CoA
CoASH
fatty acids
+ NH4+
NADH
NAD+
Ala
Citric acid cycle
1. citrate synthase
Oxaloacetate
Citrate
2. aconitase
cis-Aconitate
Isocitrate
3. isocitrate dehydrogenase
Oxalosuccinate
α-Ketoglutarate
4. α-ketoglutarate dehydrogenase
Succinyl-CoA
5. succinyl-CoA synthetase
Succinate
6. succinate dehydrogenase
FADH2
FAD
Fumarate
7. fumarase
L-Malate
8. malate dehydrogenase
Asp
Glu





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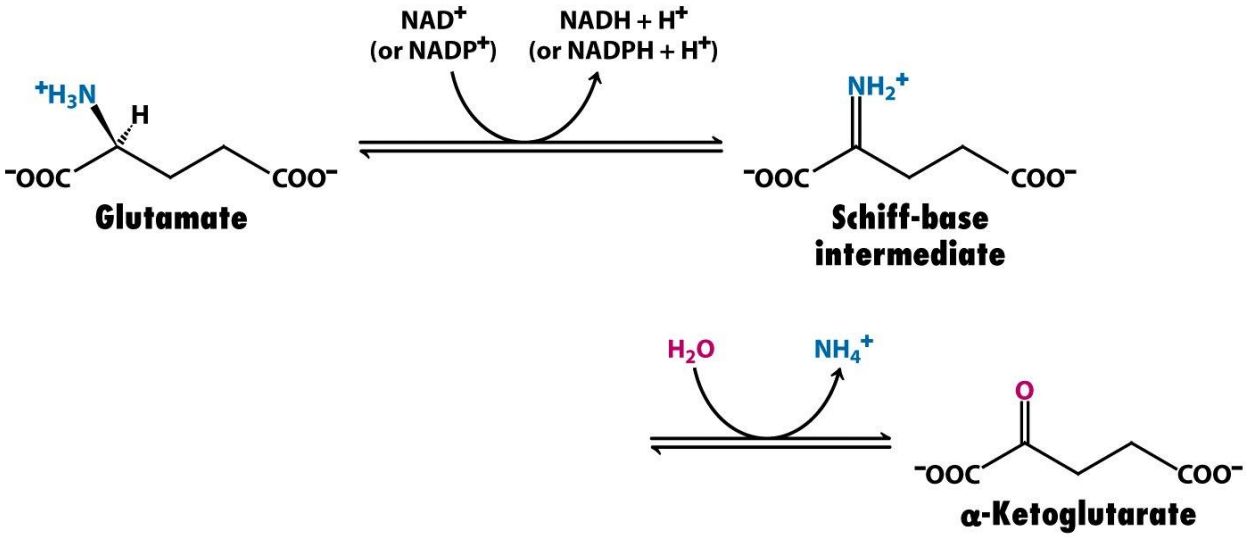

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